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REINFORCEMENT LEARNING

Teaching towards better decisions!

- ▶ In **supervised learning**, the data collection process involves collecting pairs of inputs and corresponding ground truth outputs. These pairs are used to train the model so that it can learn the mapping from inputs to outputs.
- ▶ In **unsupervised learning**, the data collection process involves collecting only the input data, as there are no corresponding ground truth outputs.
- ▶ In **reinforcement learning**, the data collection process involves the agent interacting with the environment and collecting states, actions, and rewards. This data is then used to train the model to learn a policy that maximizes the expected cumulative reward.



DATA COLLECTION



TRAINING

Data Collection in RL

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REINFORCEMENT LEARNING

Teaching through rewards for better decisions!



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Data Collection in RL

WHAT ELSE IN THIS COURSE?

Deep learning is a powerful tool that uses neural networks with many layers (hence “deep”) and can be applied to various types of learning tasks, including supervised, unsupervised, and reinforcement learning.

A neural network is a type of computer algorithm inspired by the structure of the human brain. It is designed to recognize patterns in data and make predictions or decisions based on it.

Feature Selection in machine learning is the process of selecting the most important features or variables from the dataset.

ML Pipeline, as a series of automated steps that streamline the entire process of creating a machine learning model, including collecting and cleaning data, selecting features, choosing and training a model, evaluating its performance, deploying it, and monitoring and updating it as needed.

